

Use Cases

Use Case 1: Autonomous Campus Guide (Audio-Instruction Mode – Navigation)

Actor

- Primary: Student, newcomer, or visitor
- Supporting: Robot assistant (speech recognition, LLM, navigation AI)
- External: Campus indoor map database

Goal

Get clear step-by-step audio navigation instructions to reach a target location (e.g., robotics lab).

Preconditions

- Robot is active and connected to the Context Broker.
- Indoor map accessible and up to date.
- Microphone, speaker, and speech pipeline functional.

Main Flow (Steps)

1. User Request

- a. Student: “How do I get to the robotics lab?”
- b. Robot captures speech → converts to text.

2. Command Interpretation

- a. LLM processes input → intent = navigation request.
- b. Queries Context Broker for location coordinates.

3. Path Calculation

- a. Navigation AI computes optimal path on indoor map.
- b. Path broken into human-understandable steps.

4. Instruction Delivery

- a. Robot (TTS): “Walk straight until you see the library, then turn left. The robotics lab is at the end of the hallway.”
- b. Optionally: Display path on LCD.

5. Interactive Clarification (optional)

- a. Student: “Repeat the last step.”

- b. Robot: repeats/explains with landmarks.

Postconditions

- Student receives clear navigation instructions.
- Interaction logged for analytics.

Alternative Flows

- Unknown location → “I don’t have the robotics lab location yet.”
- Noise issue → “Can you repeat your request?”

Use Case 2: Campus FAQ & General Interaction

Actor

- Primary: Student, newcomer, or visitor
- Supporting: Robot assistant (speech recognition, LLM, FAQ knowledge base)
- External: Campus knowledge base (FAQs, events, news)

Goal

Provide general campus-related information and engage in light, conversational dialogue.

Preconditions

- Robot is active and online.
- FAQ database and campus calendar accessible.
- Speech-to-text and text-to-speech functional.

Main Flow (Examples)

1. Greetings & Small Talk

- a. Robot: “Hello! Welcome to campus. How can I help you today?”

2. Campus FAQs

- a. Student: “Where is the cafeteria?”
- b. Robot: “The cafeteria is in Building A, next to the main hall. It’s open until 6 PM.”

3. Events & News

- a. Student: “What events are happening today?”

- b. Robot: "The robotics club workshop starts at 2 PM in Building B."

4. Well-being Support

- a. Student: "I'm stressed about exams."
- b. Robot: "That's normal. You can visit the counseling center in Building C."

5. Contextual Curiosity

- a. Student: "What's inside the engineering building?"
- b. Robot: "It hosts the robotics lab, electronics workshop, and the AI research center."

Postconditions

- Student gets useful info and feels engaged.
- Robot seen as a friendly assistant.

Alternative Flows

- Unknown question → "I don't know yet, but you can check at the info desk."
- Noise issue → "Sorry, can you repeat that?"